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Course: Software Testing and Quality Assurance Laboratory	Code: BIT26PE11
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Assignment 2: Write a test case using white box testing for the given code. 1. Calculate Cyclomatic complexity 2. Control flow testing 3. Data flow testing	

1. THEORETICAL BACKGROUND

1.1 White Box Testing Approach

White box testing (also known as clear box, glass box, or structural testing) is a software testing methodology that focuses on the **internal logic and structure** of the code. Unlike black box testing, the tester has access to the source code and designs test cases based on its implementation.

- **Primary Goal:** To ensure that all internal components of a program (statements, branches, and paths) are working properly.
- **Common Focus:** Structural elements such as decision points, loops, and data movements.
- **Quantitative Objective:** It allows testers to measure the "completeness" of their testing through coverage metrics.

1.2 Test Adequacy and Coverage Criteria

Test adequacy criteria act as a guide for testers to determine if the current set of test data is sufficient to stop testing.

- **Statement Adequacy:** A test set is considered statement-adequate if it causes every statement in the program to be executed at least once.
- **Branch/Decision Adequacy:** A test set is adequate if it causes every branch (True and False outcomes of a decision) to be executed.
- **Test Coverage:** This is the measured degree to which the code has been exercised. Coverage can often be less than 100% due to unreachable code, lack of resources (time/tools), or project constraints.

2. WHITE BOX TESTING TECHNIQUES

2.1 Control Flow Graph (CFG)

A Control Flow Graph is a graphical representation of all possible paths that might be traversed through a program during execution.

- **Nodes:** Represent sequential statements or blocks of code. Decisions and looping predicates are always represented as distinct nodes.
- **Edges:** Represent the transfer of control from one node to another. The direction depends on the outcome of a predicate (True or False).
- **Basic Blocks:** Sequential statements are often combined into a single block; if the first statement in the block executes, all others in that block will also execute.

2.2 Cyclomatic Complexity

Cyclomatic complexity is a software metric used to indicate the complexity of a program. It directly measures the number of linearly independent paths through a program's source code.

The Formula:

$$V(G) = E - N + 2$$

- **E:** Number of edges in the flow graph.
- **N:** Number of nodes in the flow graph.
- **Independent Path:** Any path that introduces at least one new edge that has not been traversed before. A set of these paths is called a **Basis Set**.

3. DATA FLOW TESTING TECHNIQUE

Data flow testing focuses on the lifecycle of variables: where they are assigned a value and where those values are used.

3.1 Definitions and Uses

- **Definition (def):** A variable is defined when a value is assigned to it (e.g., $Y = 26 * X$ defines Y).
- **Use:** A variable is used when its value is utilized but not changed.
 - **Predicate Use (p-use):** The variable is used in a decision or condition (e.g., if $(X > 98)$).
 - **Computational Use (c-use):** The variable is used in a calculation (e.g., $Y = 26 * X$).

3.2 Def-Use Paths

A **def-use path** is a path from a variable definition to a statement where that specific definition is used. To satisfy the **All-Def-Use Criterion**, the tester must:

1. Identify and classify all occurrences of variables.
2. Generate test data to exercise all pairs of definitions and their corresponding uses.

4. LOOP TESTING

Loop testing is a white box technique that focuses exclusively on the validity of loop constructs. Test cases are designed to exercise the loop at its boundaries:

- **Simple Loops:** Skip the loop, pass once, pass twice, pass m times .
- **Nested Loops:** Test the innermost loop while holding outer loops at minimum values.
- **Concatenated Loops:** Test independently if they are unrelated.

CONCLUSION

White Box testing provides a rigorous mathematical approach to software verification. By calculating **Cyclomatic Complexity**, developers can determine the minimum number of test cases required for full path coverage. Furthermore, **Data Flow Testing** ensures that variables are correctly initialized and utilized throughout the execution lifecycle, significantly reducing the probability of logical errors and data anomalies.